A. Separate tables (recommended)

Table 1. Reweighting granularity.

Inter-chunk	Intra-chunk	RoboTwin 2.0	RoboDojo
—	—	0.0	0.0
✓	—	0.0	0.0
—	✓	0.0	0.0
✓	✓	0.0	0.0

Table 2. Entropy-based weight assignment.

Weight assignment	RoboTwin 2.0	RoboDojo
Uniform	0.0	0.0
Shuffled	0.0	0.0
Low-entropy	0.0	0.0
High-entropy (UACA)	0.0	0.0

Table 3. Failure-tail handling.

Variant	RoboTwin 2.0	RoboDojo
GRPO	0.0	0.0
UACA w/o tail handling	0.0	0.0
UACA	0.0	0.0

All values are placeholders. Tables report success rate (%).

B. Grouped table (alternative)

Table 4. Grouped ablation results.

Variant	RoboTwin 2.0	RoboDojo
Reweighting granularity		
Uniform	0.0	0.0
Inter-chunk only	0.0	0.0
Intra-chunk only	0.0	0.0
Inter- and intra-chunk	0.0	0.0
Weight assignment		
Shuffled	0.0	0.0
Low-entropy	0.0	0.0
High-entropy	0.0	0.0
Failure-tail handling (GRPO)		
GRPO	0.0	0.0
UACA w/o tail handling	0.0	0.0
UACA	0.0	0.0

Supplementary configurations and weighting rules

Table 5. Denoising and execution settings.

Setting	Value	RoboTwin 2.0		RoboDojo	
		Orig.	+ UACA	Orig.	+ UACA
Denoising steps N	N_{low}	0.0	0.0	0.0	0.0
	N_0	0.0	0.0	0.0	0.0
	N_{high}	0.0	0.0	0.0	0.0
Execution horizon C	C_{low}	0.0	0.0	0.0	0.0
	C_0	0.0	0.0	0.0	0.0
	C_{high}	0.0	0.0	0.0	0.0

Weight mappings and estimation samples

Table 6. Entropy-to-weight mappings.

Weight mapping	RoboTwin 2.0	RoboDojo
Uniform	0.0	0.0
Hard top-q	0.0	0.0
Rank-linear	0.0	0.0
Sigmoid	0.0	0.0
Normalized exponential	0.0	0.0

Table 7. Samples used for entropy estimation.

K	RoboTwin 2.0	RoboDojo	Time (ms)
2	0.0	0.0	0.0
4	0.0	0.0	0.0
8	0.0	0.0	0.0
16	0.0	0.0	0.0

Subscript 0 denotes the default setting.

Weighting hyperparameters

Table 8. Reweighting hyperparameters.

Parameter	Value	RoboTwin 2.0	RoboDojo
τ_w	$0.5\tau_0$	0.0	0.0
	τ_0	0.0	0.0
	$2\tau_0$	0.0	0.0
η	$0.5\eta_0$	0.0	0.0
	η_0	0.0	0.0
	$\min(2\eta_0, 1)$	0.0	0.0

Additional component studies

Table 9. Reweighted policy objectives.

Q term	Supervised loss	RoboTwin 2.0	RoboDojo
—	—	0.0	0.0
✓	—	0.0	0.0
—	✓	0.0	0.0
✓	✓	0.0	0.0

Table 10. Reweighting annealing and dropout.

Annealing	Dropout	RoboTwin 2.0	RoboDojo
—	—	0.0	0.0
✓	—	0.0	0.0
—	✓	0.0	0.0
✓	✓	0.0	0.0

Table 11. Reweighting positive and negative advantages.

Variant	RoboTwin 2.0	RoboDojo
GRPO	0.0	0.0
Positive only	0.0	0.0
Negative only	0.0	0.0
Both	0.0	0.0

Requested main-table revision

Table 12. xxx

Method	RoboTwin 2.0		RoboDojo	
	Orig.	+ UACA	Orig.	+ UACA
Base policies				
X-VLA	0.0	—	0.0	—
Motus	0.0	—	0.0	—
VPP2	0.0	—	0.0	—
Online post-training				
π_0	0.0	—	0.0	—
+ GRPO	0.0	0.0	0.0	0.0
+ PPO	0.0	0.0	0.0	0.0
$\pi_{0.5}$	0.0	—	0.0	—
+ GRPO	0.0	0.0	0.0	0.0
+ PPO	0.0	0.0	0.0	0.0
Fast-WAM	0.0	—	0.0	—
+ GRPO	0.0	0.0	0.0	0.0
+ PPO	0.0	0.0	0.0	0.0
OpenWAM	0.0	—	0.0	—
+ GRPO	0.0	0.0	0.0	0.0
+ PPO	0.0	0.0	0.0	0.0

Interface summary for the appendix

Table 13. Training signals and UACA variants.

Method	Reweighted term	Variant	Granularity
GRPO / PPO	Advantage	UACA-Adv	Action
DSRL	Actor value term	UACA-Q	Chunk
RL Token	Actor value term	UACA-Q	Chunk
	Action regression	UACA-Sup	Action
EXPO-FT	Actor value term	UACA-Q	Chunk
	Flow-matching loss	UACA-Sup	Action
Filtered BC	Flow-matching loss	UACA-Sup	Action